

Exploring Deep Reservoir Computing with structured modular architectures

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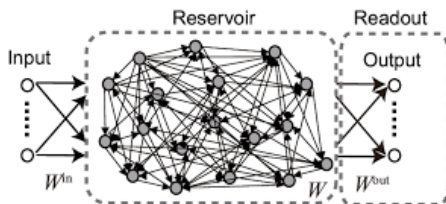
Research question:

"What is the impact of structured modularities on long short-term memory and learning in reservoir based architectures?"

Intro to Reservoir Computing

Reservoir Computing (RC) is a paradigm for efficient training of Recurrent Neural Networks (RNNs).

It works as a **dynamical system** where the reservoir maps inputs into a high-dimensional state space. At the same time keeps the Markovian bias of RNNs (ESP).



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¹Image from here

Leaky Echo State Networks

Leaky Echo State Networks (ESNs) have a *fixed, randomly connected* reservoir.

- Only the readout is trained (linear regression), ensuring efficiency.

Mathematical formulation of reservoir state update:

$$h(t) = (1 - \alpha)h(t - 1) + \alpha \tanh(W_{in}u(t) + W_{rec}h(t - 1) + b)$$

where W_{in}/W_{rec} are input/recurrent weights, $u(t)$ is input, and b is bias.

Readout Layer

The readout layer is a simple linear layer that maps the reservoir state to the output:

$$y(t) = W_{out}h(t)$$

It can be easily trained in closed form using ridge regression:

$$W_{out} = (H^T H + \lambda I)^{-1} H^T Y$$

where H is the matrix of reservoir states, Y is the matrix of target outputs, and λ is a regularization parameter.

Echo State Property (ESP)

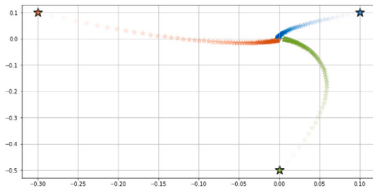
For stable dynamics and temporal dependency capture, the **ESP** must hold for any starting pair $h_1(0), h_2(0)$:

$$\lim_{t \rightarrow \infty} \|h_1(t) - h_2(t)\| = 0$$

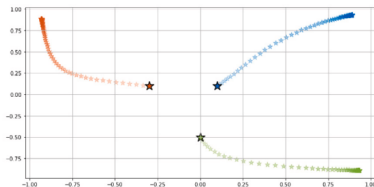
- The reservoir state is uniquely determined by input history, independent of initialization.
- Condition: Stability $\rho(W_{\text{rec}}) < 1$ and contractivity $\sigma_{\max}(W_{\text{rec}}) < 1$.

$$\rho(A) \leq \sigma_{\max}(A).$$

ESN behave as "*fading memory*" filter over past inputs.



(a) ESN with ESP.



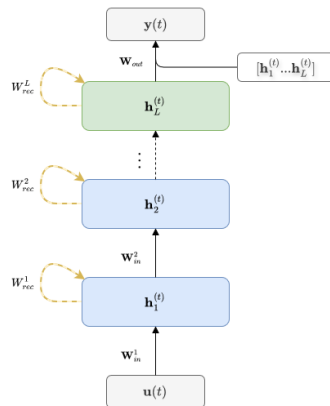
(b) ESN without ESP.

Deep Echo State Networks

Adding **depth** to the reservoir enhances representation capabilities across different time scales, improving performance on complex temporal tasks.

Deep Echo State Networks (DeepESNs):

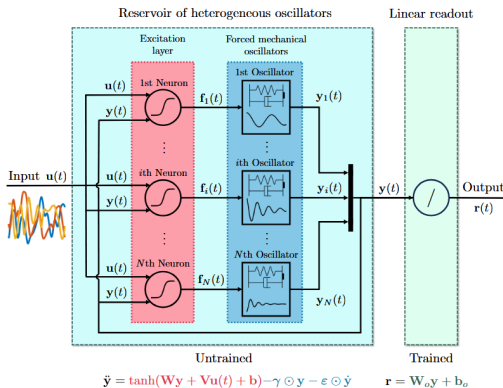
- Progressively filtering of input signal across layers
- Preserve ESP over layers



Stacked Reservoir Architecture

Recurrent Oscillatory Networks

Recurrent Oscillatory Networks (RONs) are a generalization of Leaky-ESNs, where the reservoir is designed to have oscillatory dynamics.



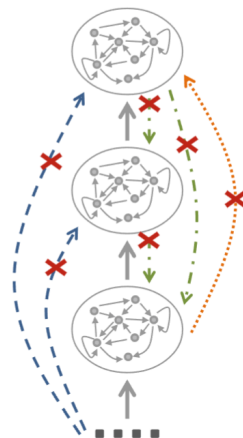
Modular Reservoir Computing

We introduce the modularity concept on DeepESNs, since they are a specific case of modular architecture.

The reservoir matrix W_{DeepESN} is block lower-triangular:

$$W_{\text{DeepESN}} = \begin{pmatrix} W_{\text{rec}}^{(1)} & 0 & \cdots & 0 \\ W_{\text{in}}^{(2)} & W_{\text{rec}}^{(2)} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & W_{\text{in}}^{(L)} & W_{\text{rec}}^{(L)} \end{pmatrix}$$

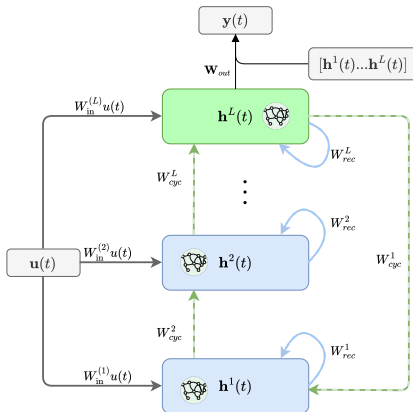
where $W_{\text{rec}}^{(l)}$ is the recurrent weight matrix for layer l , and $W_{\text{in}}^{(l)}$ is the inter-layer weight matrix connecting layer $l - 1$ to layer l .



DeepESN as constrained version of shallow reservoir

Modular Cycle ESN

The **MC-ESN**, present modules connected in a cycle, and the last layer is connected back to the first one.



MC-ESN modularity

The reservoir matrix $W_{\text{MC-ESN}}$ is given by:

$$W_{\text{MC-ESN}} = \begin{pmatrix} W_{\text{rec}}^{(1)} & 0 & \dots & W_{\text{cyc}}^{(1)} \\ W_{\text{cyc}}^{(2)} & W_{\text{rec}}^{(2)} & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & W_{\text{cyc}}^{(L)} & W_{\text{rec}}^{(L)} \end{pmatrix}$$

And the equation for the state update is:

$$\begin{aligned} h^{(1)}(t) &= (1 - \alpha) h^{(1)}(t - 1) \\ &\quad + \alpha \tanh \left(W_{\text{in}}^{(1)} u(t) + W_{\text{rec}}^{(1)} h^{(1)}(t - 1) + W_{\text{cyc}}^{(1)} h^{(L)}(t - 1) \right), \\ h^{(l)}(t) &= (1 - \alpha) h^{(l)}(t - 1) \\ &\quad + \alpha \tanh \left(W_{\text{in}}^{(l)} u(t) + W_{\text{rec}}^{(l)} h^{(l)}(t - 1) + W_{\text{cyc}}^{(l)} h^{(l-1)}(t - 1) \right). \end{aligned} \tag{1}$$

Modular Antisymmetric ESN

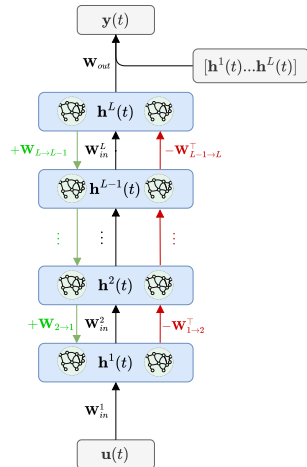
Another approach is the **MA-ESN**, with antisymmetric modules across layers..

We add a constraint on the reservoir weight matrix to enforce antisymmetry:

$$\text{Constraint: } C^{(l)} = -C^{(l-1)\top} \text{ }^a$$

The reservoir matrix $W_{\text{MA-ESN}}$ is given by:

^a $C^{(l)}$ represents where the signal is projected, in this case from if is the subdiagonal is forwarding toward l from $l-1$, the other way around for upper diagonal skew matrices.



MA-ESN modularity

The reservoir matrix $W_{\text{MA-ESN}}$ is given by:

$$W_{\text{MA-ESN}} = \begin{pmatrix} W_{\text{rec}}^{(1)} - \gamma I & -\epsilon C^{(1)\top} & 0 & \dots & 0 \\ \epsilon C^{(2)} & W_{\text{rec}}^{(2)} - \gamma I & -\epsilon C^{(2)\top} & \dots & 0 \\ 0 & \epsilon C^{(3)} & W_{\text{rec}}^{(3)} - \gamma I & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\epsilon C^{(L-1)\top} \\ 0 & \dots & 0 & \epsilon C^{(L)} & W_{\text{rec}}^{(L)} - \gamma I \end{pmatrix}$$

The state update equations for the Modular Antisymmetric ESN are:

$$\begin{aligned} h^{(l)}(t) = (1 - \alpha)h^{(l)}(t-1) + \alpha \tanh \bigg(& \underbrace{W_{\text{in}}^{(l)} h^{(l-1)}(t)}_{\text{Current Input}} \\ & + \underbrace{(W_{\text{rec}}^{(l)} - \gamma I) h^{(l)}(t-1)}_{\text{Local Memory}} \\ & + \underbrace{\epsilon \left(C^{(l)} h^{(l-1)}(t-1) - C^{\top(l)} h^{(l+1)}(t-1) \right)}_{\text{Antisymmetric Coupling}} \bigg) \end{aligned} \quad (2)$$

Memory Capacity

Memory Capacity (MC) ⁴ is a measure of how well a RNNs can recall input signal from the past.

Is defined as:

$$MC = \sum_{k=1}^{\infty} \text{corr}^2(y(t-k), \hat{y}(t-k))$$

where $y(t-k)$ is the actual output at time $t-k$ and $\hat{y}(t-k)$ is the output predicted by the network based on its current state.

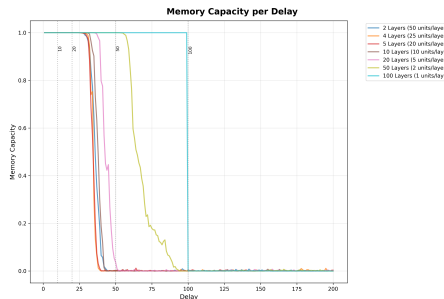
⁴Jaeger, "Short Term Memory in Echo State Networks"

Memory Capacity results

MC-ESN can recover Simple Cycle Reservoirs (SCR)⁵ behaviour, which are known to have maximal $MC \leq N$ where N is the number of neurons in a linear reservoir.

TABLE 4.1: Memory Capacity results for different architectures averaged over 3 trials from the best performing model. The higher the better.

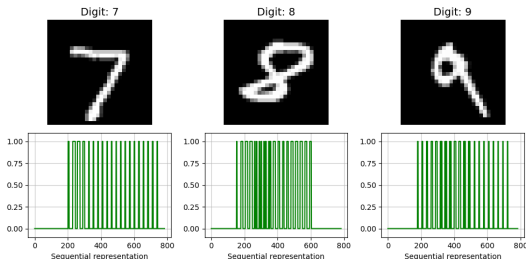
Architecture	Train MC	Test MC
DeepESN	60.3 ± 2.64	56.01 ± 2.45
MC-ESN	99.1 ± 0.1	99 ± 0.1
MA-ESN	60.51 ± 2.16	57.83 ± 2.8
DeepRON	50.58 ± 2.2	47.55 ± 3.1
MC-RON	41.9 ± 2.13	38.86 ± 2.52
MA-RON	53.35 ± 2.16	52.10 ± 1.8



⁵Rodan and Tino, "Minimum Complexity Echo State Network"

Long short-term memory

We evaluate the long short-term memory capabilities of the proposed architectures using progressively complex tasks on the MNIST dataset, pixel by pixel, pixel permutation and noise-padded



Results on long short-term memory

Modular Antisymmetric showed improvement in long-term memorization tasks

Architecture	sMNIST $\uparrow$	psMNIST $\uparrow$	npCIFAR-10 $\uparrow$
ESN	82.93% $\pm$ 0.89	78.87% $\pm$ 1.82	32.06% $\pm$ 0.19
DeepESN	85.94% $\pm$ 0.46	80.62% $\pm$ 0.95	<u>39.83%</u> $\pm$ 0.19
MC-ESN	86.81% $\pm$ 0.49	75.92% $\pm$ 0.90	32.28% $\pm$ 0.19
MC-ESN (No Recurrence)	86.32% $\pm$ 0.35	79.31% $\pm$ 1.62	33.91% $\pm$ 0.39
MA-ESN	<u>88.29%</u> $\pm$ 0.89	<u>81.33%</u> $\pm$ 0.65	34.37% $\pm$ 0.16
RON	<u>90.7%</u> $\pm$ 0.51	<u>86.5%</u> $\pm$ 1.23	28.42% $\pm$ 0.21
DeepRON	84.8% $\pm$ 1.4	81.6% $\pm$ 1.3	26.82% $\pm$ 0.1
MC-RON	89.17% $\pm$ 0.8	51.03% $\pm$ 0.8	26.36% $\pm$ 0.23
MC-RON (No Recurrence)	61.25% $\pm$ 0.51	39.6% $\pm$ 1.4	26.87% $\pm$ 0.12
MA-RON	88.7% $\pm$ 0.62	83.32% $\pm$ 2.1	<u>32.12%</u> $\pm$ 0.15

Results on long short-term memory

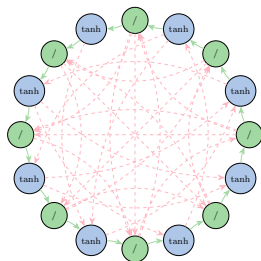
Architecture	FordA $\uparrow$	FordB $\uparrow$
ESN	$67.5\% \pm 0.2$	$58.2\% \pm 0.4$
DeepESN	$71.2\% \pm 0.16$	$62.9\% \pm 0.3$
MC-ESN	$69.8\% \pm 0.3$	$63.3\% \pm 0.2$
MC-ESN (No Recurrence)	$70.5\% \pm 0.3$	$62.9\% \pm 0.2$
MA-ESN	<u>72.3 ± 0.24</u>	<u>$63.7\% \pm 0.1$</u>
RON	<u>$71\% \pm 0.2$</u>	$64\% \pm 0.7$
DeepRON	$66.3\% \pm 1.1$	<u>$69\% \pm 1.3$</u>
MC-RON	$65.2\% \pm 0.6$	$67.1\% \pm 0.3$
MC-RON (No Recurrence)	$55.5\% \pm 0.03$	$57.1\% \pm 0.01$
MA-RON	$69\% \pm 0.02$	$67.8\% \pm 0.8$

Forecasting

Architecture	MG84 NRMSE ↓
ESN	$3 \times 10^{-2} \pm 2.4 \times 10^{-3}$
DeepESN	$\underline{1.2 \times 10^{-2} \pm 4 \times 10^{-3}}$
MC-ESN	$2.3 \times 10^{-2} \pm 4 \times 10^{-3}$
MC-ESN (No Recurrence)	$5.1 \times 10^{-2} \pm 4 \times 10^{-3}$
MA-ESN	$2 \times 10^{-2} \pm 1 \times 10^{-2}$
RON	$\underline{1.8 \times 10^{-2} \pm 6 \times 10^{-3}}$
DeepRON	$2.2 \times 10^{-2} \pm 5 \times 10^{-3}$
MC-RON	$4 \times 10^{-1} \pm 2 \times 10^{-3}$
MC-RON (No Recurrence)	$4.5 \times 10^{-1} \pm 3.6 \times 10^{-3}$
MA-RON	$1.8 \times 10^{-1} \pm 3.3 \times 10^{-2}$

Wrap-up and future work

- Modular architectures can enhance memory and learning capabilities of reservoir computing models.
- MC-ESN can achieve maximal memory capacity, while MA-ESN shows improved performance on long-term dependencies.
- **Future work:** Eterogeneous layers, more complex modular topologies, small-world and scale-free.



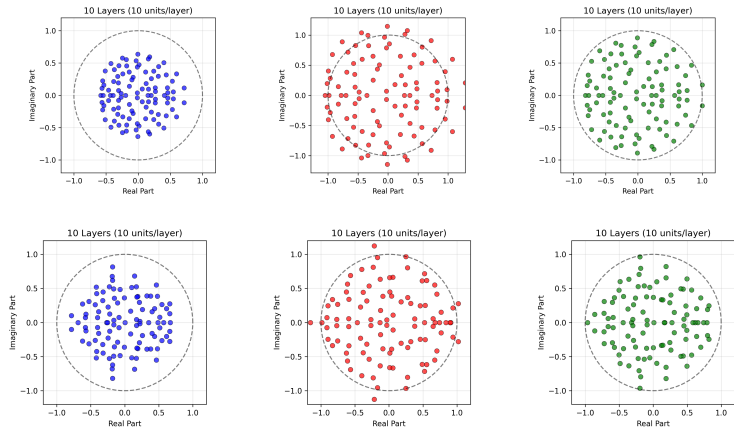
References I

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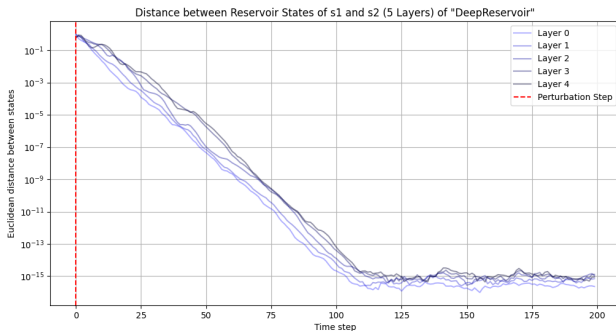
Thank you!

Supporting Materials: Eigenvalues distributions MC-ESN and MA-ESN

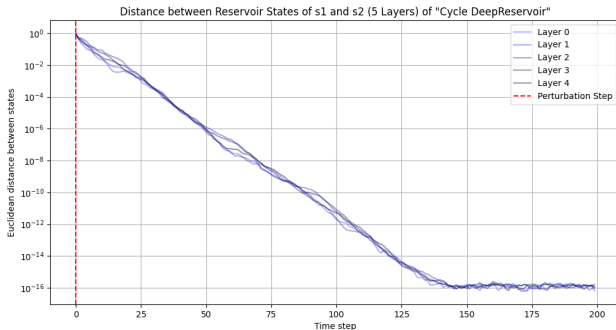
We present three settings: a contractive, one showing violation of ESP and last showing edge of stability.



Supporting Materials: Reservoir states differences (DeepESN)



Supporting Materials: Reservoir states differences (MC-ESN)



Supporting Materials: Reservoir states differences (MA-ESN)

